Arthropod-Plant Interactions
https://doi.org/10.1007/s11829-020-09770-9

ORIGINAL PAPER

Beyond robbery: the role of upside-down behaviour performed by small oil-collecting bees in Malpighiaceae

Gudryan J. Baronio[1] · Vinícius Lourenço Garcia de Brito[2] · Anselmo Nogueira[3] · André Rodrigo Rech[1,4]

Received: 27 September 2019 / Accepted: 25 May 2020
© Springer Nature B.V. 2020

Abstract

In plant–pollination interactions, the antagonists collect the same set of resources offered to legitimate visitors, but their morphologies and behavioural approaches possibly make their effect on plant fitness negative rather than positive. Depending on the morphology of Malpighiaceae flowers, these small-bodied bees can adopt an upside-down position, biting and damaging the floral pedicel. Therefore, we hypothesized that small-bodied bees may reduce female reproductive fitness, due to increased floral abscission. We observed small *Tetrapedia* bees approaching bagged and unbagged flowers of *Peixotoa tomentosa*, and compared abscission rates associated with the upside-down behavior performance. When *Tetrapedia* bees exhibited the upside-down behaviour the floral abscission was about 1.7 times more frequent than without this behaviour. We observed that abscission frequency increased to 80% if upside-down behavior is performed at least three times per flower. Additionally, the frequency of upside-down behaviours were positively related to the number of flowers visited per plant. Overall, the upside-down behaviour of *Tetrapedia* compromised most flowers that were viable for fruit formation, potentially decreasing the female reproductive fitness of plant population. We emphasize that different roles of floral visitors, including the robbers, are associated with visitors' body size and behaviour. Such features of interactions determine whether floral visitors are pollinators, thieves or robbers and shape a continuum from mutualistic to antagonistic interactions.

Keywords Behavioural constancy · Floral abscission · Mutualism-antagonism continuum · Pollination · *Tetrapedia*

Introduction

In natural communities, there is a continuum of interspecific interactions, ranging from mutualisms to antagonisms, which is modulated by differences in morphological and behavioural traits of the species involved (Thompson 2005; Irwin et al. 2015; Quinalha et al. 2017). Due to their diffuse and mostly facultative nature, plant-pollinator interactions are an excellent study system to investigate how animals can change their behaviour to perform different tasks and meet priorities during foraging, as well as how plants respond to such modulation. These interactions may result in strong direct and indirect effects on plant fitness varying from positive to negative outcomes if costs exceed benefits or vice-versa (Irwin et al. 2015; Guimarães et al. 2017). In general, pollinators and antagonists that visit flowers collect the same set of resources, but the morphology and behaviour of antagonists that feed on flowers impose costs without proportional benefits for those flowers, making the net effect of their actions negative rather than positive for plants (Irwin et al. 2010; Solís-Montero et al. 2015).

Handling Editor: Christina Mogren.

✉ Gudryan J. Baronio
gudryan@gmail.com

[1] Programa de Pós-Graduação em Ciência Florestal, Universidade Federal dos Vales do Jequitinhonha e Mucuri, Campus JK, Rodovia MGT 367 - Km 583, Nº 5.000, Alto da Jacuba, Diamantina, MG CEP 39100-000, Brazil

[2] Instituto de Biologia, Universidade Federal de Uberlândia, Campus Umuarama, Uberlândia, Minas Gerais CEP 38400-902, Brazil

[3] Centro de Ciências Naturais e Humanas, Universidade Federal do ABC, São Bernardo do Campo, SP CEP 09606-070, Brazil

[4] Faculdade Interdisciplinar em Humanidades, Universidade Federal dos Vales do Jequitinhonha e Mucuri, Campus JK, Rodovia MGT 367 - Km 583, Nº 5.000, Alto da Jacuba, Diamantina, MG CEP 39100-000, Brazil

Published online: 08 June 2020

Oil-collecting bees (e.g. Apidae: Centrindini, Tapinotaspidini, Tetrapedini, Ctenoplectrini and some Meliponini) use oil produced by specific floral glands (elaiophores) for brood cell construction and provisioning (Vogel 1974; Alves-dos-Santos et al. 2007). These bees have differentiated bristles on hairy legs where they collect, transport and store the oil resources (Vogel 1974; Rüttgers et al. 2015; Aguiar et al. 2017). There is a wide variation in body size among oil-collecting bee species, which determines the coupling capacity of bees to flowers (Vogel 1988; Alves dos Santos et al. 2007). Oil-collecting species of the genus *Centris* and *Epicharis* (Martins et al. 2014; previously referred as Centridini bees Martins and Melo 2016) are the main pollinators of Neotropical Malpighiaceae due to two main traits: (i) their large body sizes (although some few Centridini are small-bodied), and (ii) the conserved behaviour of large-bodied ones of completely embracing flowers and touching their reproductive structures (Fig. 1; Vogel 1974; Sigrist and Sazima 2004; Alves dos Santos et al. 2007). In order to collect oil, these large-bodied bees hold the flag petal with their mandibles and scrape the calyx oil glands with their first two sets of legs, while successfully pollinating the flowers during oil collection due to their morphological fit (Vogel 1974; Sigrist and Sazima 2004). Besides contacting floral reproductive organs, large-bodied bees also visit more flowers and spend less time per visit in comparison to smaller bee species (Sigrist and Sazima 2004; Barônio and Torezan-Silingardi 2017).

However, smaller bees also pollinate oil-rewarding flowers, while collecting pollen, as well as while collect oil from small-flowered species (Vogel 1988). These bees rarely succeed in pollinating large-flowered species as they are not able to embrace large flowers (Vogel 1974; Sigrist and Sazima 2004). There are at least two oil-collecting behaviours of smaller bees in Malpighiaceae flowers. In the first one, floral theft starts with bees landing behind the flower and scraping each pair of elaiophores one by one to collect oil without causing any floral damage (Sazima and Sazima 1989; Alves-dos-Santos et al. 2007; Sazan et al. 2014). Consequently, they collect oil by directly accessing the oil glands underneath the flower (Fig. 1), never touching the stigma (Sigrist and Sazima 2004; Cappellari et al. 2012). Legitimate visits can occur when small-bodied bee species explore pollen resources above the flower and next to the stigma, occasionally transferring pollen grains (Vogel 1988; Sigrist and Sazima 2004; Martins et al. 2013; but see Barônio and Torezan-Silingardi 2017). In the second behaviour, small-bodied bees land on the flower in an upside-down position and bite the floral pedicel which allows them to move around while scraping elaiophores (Sazima and Sazima 1989; Cappellari et al. 2012). One consequence of the biting behaviour during upside-down visits observed in the field, was that some flowers develop visible necrosis on the floral pedicel, which can evolve to floral abscission. Therefore, we expect the upside-down visit and biting behaviour to negatively affect flowers, and consequently the plant's female fitness.

Focusing on the interaction between small-bodied bees of *Tetrapedia* spp. and oil-rewarding Malpighiaceae *Peixotoa tomentosa* A. Juss. flowers as a model, we hypothesized that small-bodied bees that visit Malpighiaceae flowers and perform upside-down behavior may reduce the female reproductive fitness, not just due to pollen thief and oil robbery, but to the upside-down behaviour and consequent floral abscission. We expected the probability of floral abscission directly increases with frequency of upside-down behaviour. Due to the different behaviours of small-bodied bees that collect oil or pollen from flowers, which can vary both qualitatively and quantitatively in a community, we also expected a positive relationship between the number of visited flowers and the upside-down behaviour performance. Although different scenarios are possible, our hypothesis would be rejected if the following statements are true: (1) small-bodied bees mostly explore pollen grains, touching stigmas during flower visitation, without damaging flowers by upside-down behaviour (to explore oil resource); (2) small-bodied bees collect oil resources without robbing and damaging floral tissues and (3) floral abscission is not related to the frequency of the upside-down behaviour of small-bodied bees.

Material and methods

Study area and plant species

We conducted the study during October 2014, in savanna areas at the State Park of Caldas Novas (PESCAN), located between Caldas Novas and Rio Quente municipalities, in Southeastern Goiás state, Brazil (17° 47′ S and 48° 40′ W, WGS84 Datum—12,000 ha) and the Ecological Reserve of Clube de Caça e Pesca Itororó (CCPIU) in Uberlândia (18° 59′ S and 48° 18′ W, WGS84 Datum—640 ha), Minas Gerais state, Brazil. Preliminary observations of visiting behaviour were carried out between June and August 2011 at CCPIU. The climate is defined as Aw, with clear distinct rainy and dry seasons, with a mean annual temperature of 23 °C and mean precipitation of 1500 mm (Alvares et al. 2013).

We focused our observations on *Peixotoa tomentosa* A. Juss. as this plant species exhibits large yellow flowers, about 3 cm in diameter (Mamede et al. 2016), which facilitates observation of small-bodied bee behaviour. Furthermore, when compared to other Malpighiaceae, *P. tomentosa* produces fewer flowers (9.5 ± 5.4 flowers per inflorescence, and 2.8 ± 0.6 inflorescences per plant) during each blooming season between July and October (Barônio et al. 2017). *Peixotoa tomentosa* has from five to eight elaiophores or oil glands per flower, and provides about 1.8 times more oil

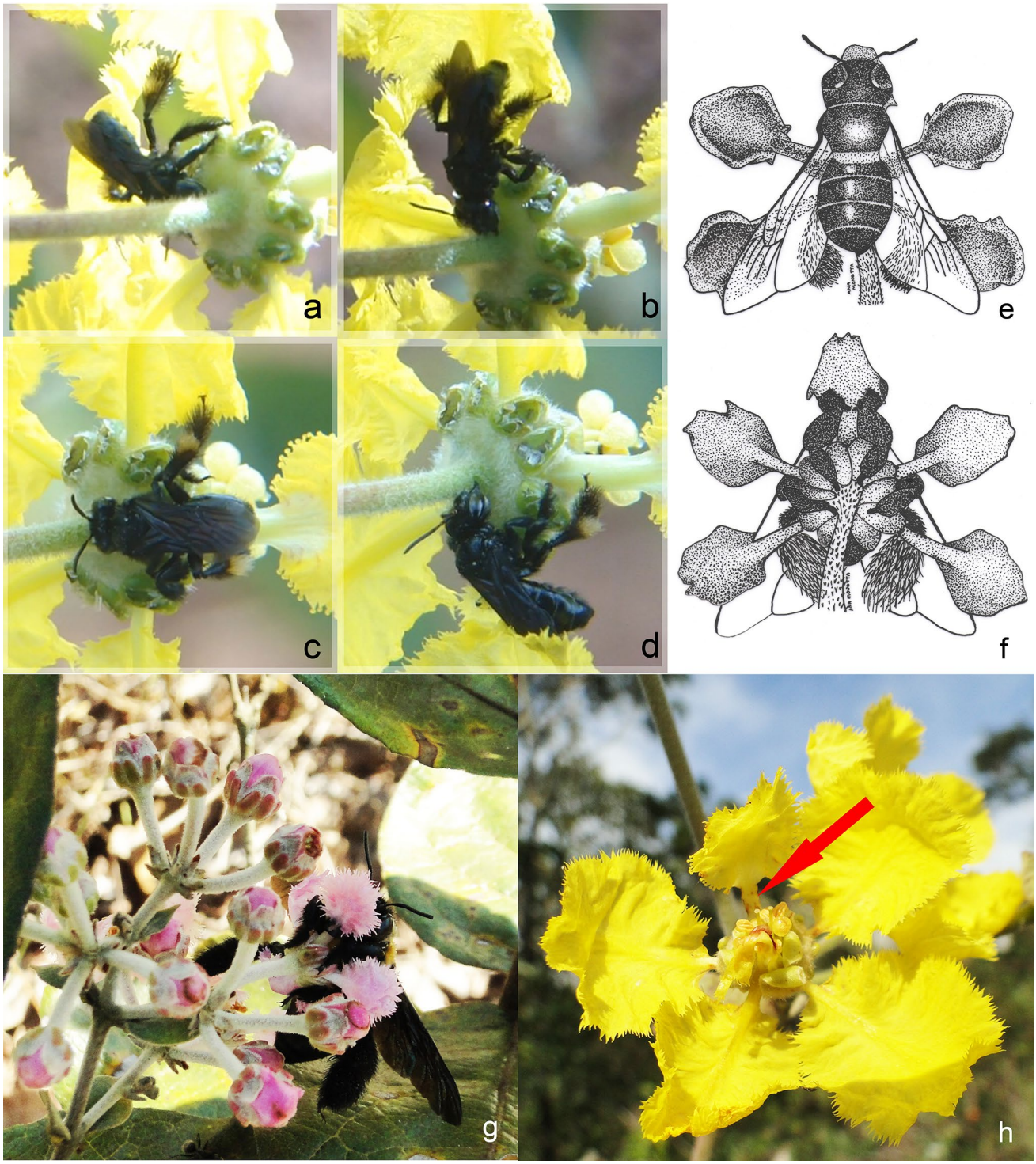


Fig. 1 Upside-down behaviour of oil collection by *Tetrapedia* (**a**–**d**). The bee lands on *P. tomentosa* flowers upside down and bites the pedicel (**a**) while rotating in a circular motion around the flower and collecting the oil from the glands with the first and second pairs of legs, depositing it onto the scopa. Note that the elaiophores have indications of previous collections of floral oil, such as ruptured cuticle, lower volume and necrosis, which may be the result of previous visits by large (*Centris* and *Epicharis*) and small-bodied bees (*Tetrapedia*). The legimate pollinating behaviour of large-bodied bees *Centris* and *Epicharis* is performed by landing on the front of the flower, touching reproductive strucutures while scrapping oil glands with the first two pairs of legs (**e**, **f**) and bitting the flag petal (red arrow). This behaviour is strongly conserved and can be seen by mandible marks left on the flag petal of Malpighiaceae flowers (**g**, **h**). Ilustrations (**e**, **f**) were made by Ana Carolina Monetta

in each flower than some sympatric species of Malpighiaceae, such as those from genus *Banisteriopsis* (Barônio and Torezan-Silingardi 2017). Data about the breeding system of *P. tomentosa* is not available, however, Neotropical Malpighiaceae species are generally characterized by limited spontaneous self-pollination due to high herkogamy and protogyny (Sigrist and Sazima 2004; Sazan et al. 2014).

Data collection

We observed the floral approaches of all bees but focused on *Tetrapedia* as it presents the upside-down behaviour that potentially damages *P. tomentosa* flowers. We observed 15 *P. tomentosa* individuals located at least 5 m away from each other. During floral observations, we did not collect bees as small movements near the plants could interfere in their behavior. However, we carefully observed floral approaches and behavior performed by bees to qualify which visits occurred with upside-down behaviour. This allowed us to distinguish if a given bee was collecting pollen or oil resources, and if they were damaging flowers according to their behavioral approach.

At our study site, two *Tetrapedia* species contributed to the oil-collecting bee guild: *T. diversipes* and *T. curvitarsis* (Barônio and Torezan-Silingardi 2017). We recorded their visits on *P. tomentosa* between 7am and 6 pm for three days (10th, 11th, and 16th of October at the PESCAN reserve, and 21th, 22th, 24th of October at the CCPIU reserve) with observational periods of 25 min per plant. Prior to each census, half the buds per shrub were bagged to observe the first visit by *Tetrapedia* bees. The second set of buds were not bagged, allowing visits by some large-bodied bees of *Centris* or *Epicharis* genus. We marked non-coloured lines (< 1 mm width) after each visit to record the frequency of upside-down behaviour. At the end of each observational period, observed flowers (visited and not visited) were bagged again to avoid other visits, and then observed for the next 10 days to record floral abscission. To confirm when large-bodied bees visited the unbagged flowers prior to visits from small-bodied bees, we observed the marks left on the flag petal by the mandibles of large-bodied bees, any broken cuticles, as well as decreased volume of elaiophores and necrosis, which are evidences of flowers visited by *Centris* or *Epicharis* bees (Vogel 1974; Sigrist and Sazima 2004). Such evidence of visitation allowed us to compare frequency of abscission with and without previous visits by large sized bees.

Statistical analysis

Using data from bagged flower observations, we performed a chi-square test to investigate the association of floral abscission in flowers that received or did not receive visits with upside-down behaviour by *Tetrapedia* bees. Additionally, we performed a logistic regression model using floral abscission (= 1) and floral retention (= 0) as response variables (Zar 2010) to determine if the probability of floral abscission was positively related to the number of visits with upside-down behaviour that each flower received during the first day of anthesis ($n = 30$). To characterize the variation in the upside-down behaviour per plant ($n = 60$ plants), we also performed a generalized linear model to test the relationship between the total number of upside-down visits per plant and number of flowers exclusively visited by *Tetrapedia* (bagged flowers). According to count data (visited flowers), we used a Poisson distribution to normalize the residuals from the model (Legendre and Legendre 1998; Crawley 2012). Finally, we used data from observations of unbagged flowers to perform another chi-square test to determine if there were different abscission proportions in flowers where *Tetrapedia* spp. performed upside-down behaviour but were or were not previously visited by large-bodied bees. All statistical analyses were performed using the package *stats* in the R software (R Core Team 2018).

Results

Tetrapedia bees visited 497 (265 bagged and 232 unbagged) flowers of *Peixotoa tomentosa* during the focal observations. Considering first flower visits, *Tetrapedia* performed upside-down behaviour in 56.6% and exclusively collected pollen in the other 43.4%. Accordingly, 75% of flowers did not fall when visited by *Tetrapedia* that did not perform the upside-down behaviour. On the other hand, 44% did fall after being visited with upside-down behaviour ($\chi^2 = 9.33$; df = 1; $p = 0.002$; Fig. 2). Thus, flower abscission after upside-down behaviour of *Tetrapedia* bees was 1.7 times greater than when bees visited flowers for pollen exclusively.

Among the unbagged flowers, large-sized bees (*Centris* or *Epicharis*) visited 56.3% of flowers, and about a third (31.8%) abscised afterwards. This proportion was not different from the unbagged flowers that abscised and were not visited by large-bodied bees (34.7–144 flowers). Thus, there is no evidence that floral abscission was related to previous visits by large-bodied bees ($\chi^2 = 0.21$; df = 1; $p = 0.65$), since we observed no difference in the proportion of flowers that were abscised with (46.6%) or without (53.1%) at least one previous large-bodied bee visit. However, we also observed small-bodied bees (*Tetrapedia*) visiting flowers depleted of oil resources. We tried to collect oil manually with capillaries after large-bodied bee visits, but there was virtually no floral oil leftover.

During all observations, flowers in each plant received from zero to eight upside-down *Tetrapedia* visits. We found a positive and exponential relationship between the number of visited flowers and the number of visits

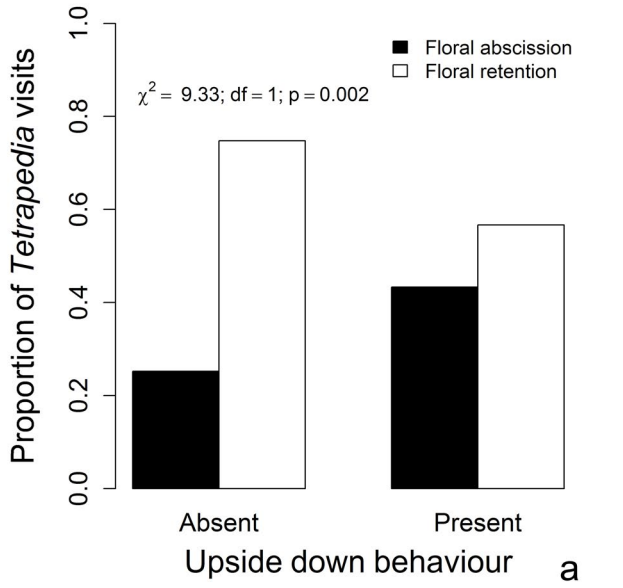




Fig. 2 Proportion of *Peixotoa tomentosa* flowers that fell after receiving upside-down visits from *Tetrapedia* bees (**a**). The upside-down behaviour occurs when bees hold themselves by biting the floral receptacle while moving along the flower and scraping the elaiophores. After *Tetrapedia* bee visits, *Peixotoa tomentosa* flowers developed necrosis in the pedicel (**b**)

exhibiting upside-down behaviour ($\chi^2 = 59.54$; df = 58; $p < 0.001$; $R^2 = 0.13$; Fig. 3). Furthermore, the probability of floral abscission increased with the frequency of upside-down (max four) visits in each flower ($\chi^2 = 18.08$; df = 28; $p < 0.001$; $R^2 = 0.44$). The flowers visited three times with upside-down behaviour had more than 80% chance of being abscised (Fig. 4). In general, the 1.7 time increase in the frequency of abscission due to upside-down

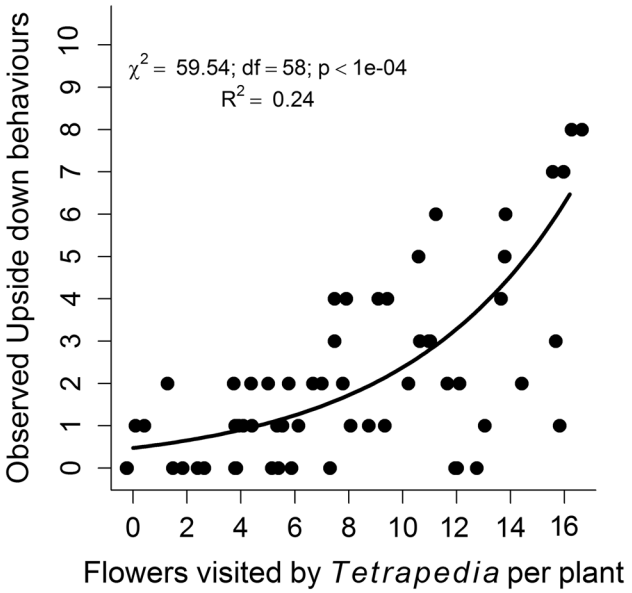


Fig. 3 General linear model (by Poisson distribution) showing the positive relationship between the number of upside-down visits and number of *Peixotoa tomentosa* flowers visited by *Tetrapedia* bees

behaviour correlated to an ~ 13 times greater probability of falling for flowers that received *Tetrapedia* upside-down visits (Odds ratio 13.4; 95% CI 2.21–81.1).

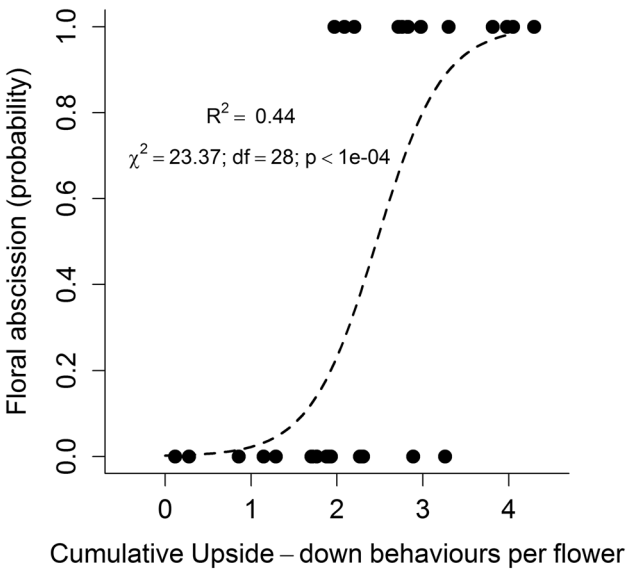


Fig. 4 Logistic model showing an increased probability of floral abscission due to sequential upside-down visits by *Tetrapedia* on the same flower of *Peixotoa tomentosa.* Due to data point overlap, the numbers above the black points show the exact occurrence of visits

Discussion

Our main results evidenced that the upside-side down behaviour of *Tetrapedia* was associated with a higher proportion of floral abscission after pedicel biting, increasing the probability of floral abscission with the accumulation of upside-down visits. Overall, the upside-down behaviour of *Tetrapedia* compromised most flowers that were viable for fruit formation, potentially decreasing the female reproductive fitness of plant population.

The upside-down behaviour has primarily been described and discussed as non-pollinating for Malpighiaceae flowers of *Banisteriopsis, Byrsonima* and *Heteropterys* species, which are mainly visited by small-bodied bees from *Tetrapedia* and *Paratetrapedia* (Sazima and Sazima 1989; Cappellari et al. 2012). To the best of our knowledge, this is the first record of upside-down behaviour on *Peixotoa tomentosa* flowers, although we have also observed the upside-down behaviour in flowers of *Banisteriopsis campestris* (A. Juss.) Little and *B. malifolia* (Nees & Mart.) B. Gates in our study areas during data collection. In contrast to the upside-down behaviour, alternative *Tetrapedia* behaviours were only adopted in a small set of flowers in *Peixotoa tomentosa*, when bees collected pollen grains and oil without damaging flowers. *Tetrapedia* visits which harvested only for oil in *P. tomentosa* flowers always involved biting the pedicel next to the sepals, as they never visited gland by gland, acting as robbers.

We found an association between floral abscission and the upside-down positioning behaviour by *Tetrapedia* bees during oil collection in *Peixotoa tomentosa*. Like large-bodied bee species, these small-bodied bees are able to bite flowers, but due to their reduced body size, they have a limited ability to embrace the entire flower and promote efficient pollination (Sazima and Sazima 1989; Sigrist and Sazima 2004; Cappellari et al. 2012; Sazan et al. 2014; Barônio and Torezan-Silingardi 2017). For example, the bee *Xanthopedia larocai* was observed walking over the elaiophores and hanging itself from the petals' pedicel using its mandibles (Sazan et al. 2014). Thus, they still reach the oil glands in the sepals, by biting the floral pedicel to access the oil resources under the flower (Sazan et al. 2014).

Consequently, small-bodied bees damaged plant tissues in the floral pedicel and increased floral abscission. Although there is a hormonal control of floral abscission, when flowers are not pollinated (van Doorn and Stead 1997; Sundaresan et al. 2015), there was no indication that floral abscission in *P. tomentosa* was due to pollination failure. This is evident as abscission even occurred in flowers that were legitimately visited by large-bodied bees. In some cases, flowers that received visits with upside-down behaviour were retained. This pattern can be explained by (i) plant mechanisms to reduce plant damage which moderate negative effects of biting behaviour and/or (ii) variable intensity of damage depending on the biting behaviour of each *Tetrapedia* specimen, which may not be strong enough to initiate floral abscission (Ascough et al. 2005; Sundaresan et al. 2015).

Small-bodied bees like *Tetrapedia* specimens are frequent visitors of Malpighiaceae, accounting for more than 20% of all bee visitors (Sazima and Sazima 1989; Barônio and Torezan-Silingardi 2017). On the other hand, large *Centris* and *Epicharis* bees, widely known as the main Malpighiaceae pollinators, are the visitors that most frequently access oil glands above the flower (Sigrist and Sazima 2004). In our study, large and small-sized bees concomitantly visited flowers of *Peixotoa tomentosa*, and over half of the flowers visited by *Tetrapedia* bees that performed the upside-down behaviour were previously visited by large-bodied bees. Therefore, floral abscission occurs even when flowers had possibly been pollinated by large-bodied bees of *Centris* and *Epicharis* genus, suggesting that *Tetrapedia* can impose a high cost when overlap mutualists and robbers (Barker and Bronstein 2016). Consequently, the upside-down behaviour of *Tetrapedia* may reduce the positive effects of large-bodied bees as pollinators if small-bodied bees visit flowers secondarily. Accordingly, the small-bodied bees that visit oil-flowers after primary visits by large-bodied bees seems to be at a disadvantage because less or no (GJB, *pers. obs.*) oil is available after larger bee harvesting. In particular, this system may contribute to the discussion about how exploitative and cooperative visitors may affect the net costs of different interactions (mutualistic or antagonistic) according to their temporal order of occurrence (see: Barker and Bronstein 2016). Furthermore, if considering all possible effects—direct and indirect—floral abscission could also reduce the floral display and consequently reduce attraction for the entire plant, not only for an individual damaged flower.

Among the small-sized bee species visiting Malpighiaceae flowers there is a highly variable range in body size that could generate—at community level—different optimal behaviours that promote the morphological fit of each bee species during flower visits (Vilas-Boas et al. 2013; Sazan et al. 2014; Barônio and Torezan-Silingardi 2017). The size-efficiency is a factor that potentially structures the floral community as the morphological and behavioural coupling phenomenon was also observed in other trait-behaviour systems, such as those between pollen flowers and buzzing bees (Mesquita-Neto et al. 2018). As large oil-collecting bees present conserved behaviour due to their body size, we expect that body size is the trait that induces bees to pollinate, steal or rob, all of which are modulated by embracing or holding flowers with mandibles or legs (Sazan et al. 2014). However,

smaller bees exhibit distinct behaviours (e.g. thief and/or robber) using their mandibles to visit the flowers and performing behaviours that may or may not cause damage, or even pollinate (Please, see: Sazan et al. 2014; Rego et al. 2018). Therefore, using their mandibles during visits to oil-flowers seems to be evolutionarily conditioned to a visitor's body size, especially since large-bodied bees are their main pollinators (Vogel 1974; Sigrist and Sazima 2004; Alves dos Santos et al. 2007). It is possible to expect that, in large-sized Malpighiaceae flowers, bee body size variation and consequent behaviour on flowers drive interactions between them to distinct outcomes—mutualism/antagonisms continuum—according to different behaviours performed (Pinto et al. 2015; Nunes et al. 2018). For instance, when flowers are large enough to prevent complete embracing by small-bodied bees like *Tetrapedia*, bee species may visit flowers and explore oil glands using the upside-down behaviour. Also, when bee species are smaller than flowers, they may visit sepals gland by gland, avoiding pedicel damage. Considering the set of behaviors performed by oil-collecting bees', we hypothesize that their interactions fall in different places in the mutualism-antagonism continuum when (i) bee body size coupling favours the reproductive fitness of a given floral morphology; or when (ii) bees that do not perfectly fit the flower to ensure efficient pollination exhibit more diffuse behaviours, such as the small-bodied oil-collecting bees visiting large Malpighiaceae flowers.

Flowers may receive better pollination services as a consequence of superior resource quantity and quality (Ramirez et al. 2011; Leonhardt and Blüthgen 2012). However, enhanced resources also increase the floral attractiveness to thieves and robbers (Adler et al. 2004; Zhang et al. 2014). Essentially, small-bodied bees of *Tetrapedia* exploited oil resources as robbers or thieves in flowers of *Peixotoa tomentosa*, and we do not know whether different floral oil composition can or cannot affect bee behavior on flowers. In our case, when *Tetrapedia* bees visited more flowers, they showed a higher frequency of upside-down positioning behaviour. Additionally, considering that flowers of *P. tomentosa* provide distinct oil quantity and chemical composition, maintaining foraging of *Tetrapedia* bees throughout a large set of visits may be advantageous for obtaining a large quantity of exclusive resources (Barônio et al. 2017). Therefore, the upside-down behavior constancy may reduce the handling costs of resource collection, allowing small-bodied bees to save time and/or energy (Bronstein et al. 2017). In addition, since the efficiency of small-bodied bees to collect oil is potentially inferior to large-bodied bees, it is hypothetical that new behavioural approaches might be evolutionarily selected when enhancing resource harvesting, supported by the size coupling between flowers and bees and behavioural constancy expectations (Sazan et al. 2014; Balfour et al. 2015; Bronstein et al. 2017).

Both small- and large-bodied bees use their mandibles to support themselves while collecting oil, especially by biting the flag petal while standing (large-bodied bees) or the floral pedicel while in upside down position (small-bodied bees). Indirectly, these two mandible's uses positively or negatively affect flowers, indicating pollination or floral abscission, respectively. In conclusion, small-bodied bees' upside-down behaviour is negative for flowers, as it is associated with floral abscission and increases the probability that it will occur according to the frequency of performed behaviors. Moreover, it may be negative for the plant populations since it could evolve from antagonist selection. Additionally, our results show that secondary visits of robbers on flowers previously visited by pollinators can cause negative effects on the exploited interactor (Barker and Bronstein 2016). Considering what determines the balance between cooperation and exploitation in mutualistic interactions (Frederickson 2013; Barker and Bronstein 2016), herein we emphasize the different roles of floral visitors, especially the occurrence of oil robbers, that are associated with body size and behaviour of *Tetrapedia* bees during visits to large *Peixotoa tomentosa* flowers. Such features of interactions determine whether floral visitors are pollinators or robbers and shape a continuum from mutualistic to antagonistic interactions in relationships between Malpighiaceae and oil-collecting bee species.

Acknowledgements This manuscript is dedicated to Stefan Vogel, who was responsible for showing us the beautiful world of interactions between bees and oil-flowers. The authors thank the staff of Parque Estadual da Serra de Caldas Novas (PESCAN) and Clube de Caça e Pesca Itororó de Uberlândia (CCPIU) for permission to access study site. This study was financed in part by the Coordenação de Aperfeiçoamento de Pessoal de Nível Superior—Brasil (CAPES)—Finance Code 001. GJB is grateful for a postdoctoral fellowship awarded by CAPES/PNPD/UFABC (process number 88882.315272/2019-01) and CAPES/PNPD/UFVJM (process number 88887.352134/2019-00). All authors are grateful for input on the manuscript by Francismeire J. Telles, Victor Aguiar de Souza Penha, Allan G. Ellis, Leonardo Galetto, and Judith Bronstein. GJB is grateful for illustration made by Ana Carolina Monetta.

References

Adler LS, Bronstein JL, Biology E (2004) Attracting antagonists: does floral nectar increase leaf herbivory? Ecology 85:1519–1526. https://doi.org/10.1890/03-0409

Aguiar CML, Caramés J, França F, Melo E (2017) Exploitation of floral resources and niche overlap within an oil-collecting bee guild (Hymenoptera, Apidae) in a neotropical savanna. Sociobiology 64:78. https://doi.org/10.13102/sociobiology.v64i1.1250

Alvares CA, Stape JL, Sentelhas PC et al (2013) Köppen's climate classification map for Brazil. Meteorol Zeitschrift 22:711–728. https://doi.org/10.1127/0941-2948/2013/0507

Alves-dos-Santos I, Machado IC, Gaglianone MC (2007) História natural das abelhas coletoras de óleo. Oecol Bras 11:544–557. https://doi.org/10.4257/oeco.2007.1104.06

Ascough GD, Nogemane N, Mtshali NP et al (2005) Flower abscission: environmental control, internal regulation and physiological responses of plants. S Afr J Bot 71:287–301. https://doi.org/10.1016/S0254-6299(15)30101-0

Balfour NJ, Gandy S, Ratnieks FLW (2015) Exploitative competition alters bee foraging and flower choice. Behav Ecol Sociobiol 69:1731–1738. https://doi.org/10.1007/s00265-015-1985-y

Barker JL, Bronstein JL (2016) Temporal Structure in cooperative interactions: what does the timing of exploitation tell us about its cost? PLoS Biol 14:e1002371. https://doi.org/10.1371/journal.pbio.1002371

Barônio GJ, Haleem MA, Marsaioli AJ, Torezan-Silingardi HM (2017) Characterization of Malpighiaceae flower-visitor interactions in a Brazilian savannah: how do floral resources and visitor abundance change over time. Flora 234:126–134. https://doi.org/10.1016/j.flora.2017.07.015

Barônio GJ, Torezan-Silingardi HM (2017) Temporal niche overlap and distinct bee ability to collect floral resources on three species of Brazilian Malpighiaceae. Apidologie 48:168–180. https://doi.org/10.1007/s13592-016-0462-6

Bronstein JL, Barker JL, Lichtenberg EM et al (2017) The behavioral ecology of nectar robbing: why be tactic constant? Curr Opin Insect Sci 21:14–18. https://doi.org/10.1016/j.cois.2017.05.013

Cappellari SC, Melo GAR, Aguiar AJC, Neff JL (2012) Floral oil collection by male *Tetrapedia* bees (Hymenoptera: Apidae: Tetrapedini). Apidologie 43:39–50. https://doi.org/10.1007/s13592-011-0072-2

Crawley MJ (2012) The R book, 2nd edn. Wiley, England

Frederickson ME (2013) Rethinking mutualism stability: cheaters and the evolution of sanctions. Q Rev Biol 88:269–295. https://doi.org/10.1086/673757

Guimarães PR, Pires MM, Jordano P et al (2017) Indirect effects drive coevolution in mutualistic networks. Nature 550:511–514. https://doi.org/10.1038/nature24273

Irwin RE, Bronstein JL, Manson JS, Richardson L (2010) Nectar Robbing: Ecological and Evolutionary Perspectives. Annu Rev Ecol Evol Syst 41:271–292. https://doi.org/10.1146/annurev.ecolsys.110308.120330

Irwin RE, Howell P, Galen C (2015) Quantifying direct vs. indirect effects of nectar robbers on male and female components of plant fitness. J Ecol 103:1487–1497. https://doi.org/10.1111/1365-2745.12476

Legendre P, Legendre L (1998) Numerical ecology. Elsevier, Amsterdam

Leonhardt SD, Blüthgen N (2012) The same, but different: pollen foraging in honeybee and bumblebee colonies. Apidologie 43:449–464. https://doi.org/10.1007/s13592-011-0112-y

Mamede MCH, Amorim AM, Sebastiani R (2016) Malpighiaceae. In: List. Espécies da Flora do Bras. http://floradobrasil.jbrj.gov.br. Accessed 10 Aug 2017

Martins AC, Melo GAR (2016) The new world oil-collecting bees *Centris* and *Epicharis* (Hymenoptera, Apidae): molecular phylogeny and biogeographic history. Zool Scr 45:22–33. https://doi.org/10.1111/zsc.12133

Martins AC, Aguiar AJC, Alves-dos-Santos I (2013) Interaction between oil-collecting bees and seven species of Plantaginaceae. Flora Morphol Distrib Funct Ecol Plants 208:401–411. https://doi.org/10.1016/j.flora.2013.07.001

Martins AC, Melo GAR, Renner SS (2014) The corbiculate bees arose from new world oil-collecting bees: implications for the origin of pollen baskets. Mol Phylogenet Evol 80:88–94. https://doi.org/10.1016/j.ympev.2014.07.003

Mesquita-Neto JN, Blüthgen N, Schlindwein C (2018) Flowers with poricidal anthers and their complex interaction networks-disentangling legitimate pollinators and illegitimate visitors. Funct Ecol 32:2321–2332. https://doi.org/10.1111/1365-2435.13204

Nunes CEP, Maruyama PK, Azevedo-Silva M, Sazima M (2018) Parasitoids turn herbivores into mutualists in a nursery system involving active pollination. Curr Biol 28:980–986.e3. https://doi.org/10.1016/j.cub.2018.02.013

Pinto CE, da Silva A, Duran Cordeiro G, Alves-dos-Santos I (2015) The body size of the oil-collecting bee *Tetrapedia diversipes* (Apidae). J Hymenopt Res 47:103–113. https://doi.org/10.3897/JHR.47.4837

Quinalha MM, Nogueira A, Ferreira G, Guimarães E (2017) Effect of mutualistic and antagonistic bees on floral resources and pollination of a savanna shrub. Flora 232:30–38. https://doi.org/10.1016/j.flora.2016.08.005

R Core Team (2018) R: A language and environment for statistical computing. R Foundation for Statistical Computing, Vienna, Austria

Ramirez SR, Eltz T, Fujiwara MK et al (2011) Asynchronous diversification in a specialized plant-pollinator mutualism. Science 333:1742–1746. https://doi.org/10.1126/science.1209175

Rego JO, Oliveira R, Jacobi CM, Schlindwein C (2018) Constant flower damage caused by a common stingless bee puts survival of a threatened buzz-pollinated species at risk. Apidologie 49:276–286. https://doi.org/10.1007/s13592-017-0552-0

Rüttgers A, Griebel M, Pastrik L et al (2015) Simulation of the oil storage process in the scopa of specialized bees. Comput Fluids 119:115–130. https://doi.org/10.1016/j.compfluid.2015.06.029

Sazan MS, Bezerra ADM, Freitas BM (2014) Oil collecting bees and *Byrsonima cydoniifolia* A. Juss. (Malpighiaceae) interactions: the prevalence of long-distance cross pollination driving reproductive success. Acad Bras Cienc 86:347–358. https://doi.org/10.1590/0001-3765201420130049

Sazima M, Sazima I (1989) Oil-gathering bees visit flowers of eglandular morphs of the oil-producing malpighiaceae. Bot Acta 102:106–111. https://doi.org/10.1111/j.1438-8677.1989.tb00073.x

Sigrist MR, Sazima M (2004) Pollination and reproductive biology of twelve species of neotropical malpighiaceae: stigma morphology and its implications for the breeding system. Ann Bot 94:33–41. https://doi.org/10.1093/aob/mch108

Solís-Montero L, Vergara CH, Vallejo-Marín M (2015) High incidence of pollen theft in natural populations of a buzz-pollinated plant. Arthropod Plant Interact 9:599–611. https://doi.org/10.1007/s11829-015-9397-5

Sundaresan S, Philosoph-Hadas S, Riov J et al (2015) Abscission of flowers and floral organs is closely associated with alkalization of the cytosol in abscission zone cells. J Exp Bot 66:1355–1368. https://doi.org/10.1093/jxb/eru483

Thompson JN (2005) The geographic mosaic of coevolution. University of Chicago Press, Chicago, p 400

van Doorn WG, Stead AD (1997) Abscission of flowers and floral parts. J Exp Bot 48:821–837. https://doi.org/10.1093/jxb/48.4.821

Vilas-Boas JC, Fava WS, Laroca S, Sigrist MR (2013) Two sympatric Byrsonima species (Malpighiaceae) differ in phenological and reproductive patterns. Flora 208:360–369. https://doi.org/10.1016/j.flora.2013.05.003

Vogel S (1974) Olblumen und olsammelnde Bienen. Trop und Subtrop Pflanzenwelt 7:285–547. https://doi.org/10.1002/fedr.19770880110

Vogel S (1988) The oil-bee oil-flower relationships—parallelism and other aspects of their evolution in space and time. Zeitschrift Fur Zool Syst und Evol 26:341–362

Zar JH (2010) Biostatistical analysis. Pearson Prentice Hall, Upper Saddle River

Zhang Y-W, Zhao J-M, Inouye DW (2014) Nectar thieves influence reproductive fitness by altering behaviour of nectar robbers and legitimate pollinators in *Corydalis ambigua* (Fumariaceae). J Ecol 102:229–237. https://doi.org/10.1111/1365-2745.12166

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.